# Prabin Pathak

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## Profile

Backend WebGIS developer experienced in designing, implementing, and maintaining large-scale GIS platforms. Proficient in Python and modern web frameworks — Django, Django REST Framework, FastAPI and Flask — and comfortable with asynchronous architectures and performance tuning. Open-source enthusiast who prioritizes clean, testable code and scalable system design.

## Employment History

2024 – Present **Lead Maintainer/Backend Developer (Tasking Manager)** – Humanitarian OpenstreetMap Team (HOT) X Naxa Pvt. Ltd.

2021 – Present **Sr. Backend Developer** – Naxa Pvt. Ltd

2019 **AWS Developer Associate Intern** – Genese Software Solution

## Education

2020 – 2022 **Masters in Computer Systems and Knowledge Engineering**
Institute of Engineering, Pulchowk Campus (TU).

2014 – 2018 **Bachelors in Computer Engineering**
Kathmandu Engineering College (TU), Kalimati, Kathmandu.

## Projects

- **Tasking Manager (HOT TM)**
  - Migrated the entire backend stack from Flask and psycopg2 to FastAPI and asyncpg, making it an efficient asynchronous application.
  - Load-tested and benchmarked the new architecture by simulating real-world load with k6, Grafana and InfluxDB to show the performance of the migrated async application.
  - Took part in moving the infrastructure from VMs to container-based services (ECS, RDS), with Terraform blue-green deployments for zero downtime.
  - Set up the test suite for the asynchronous framework and refactored the unit and integration tests.
  - Released a sandbox environment for Tasking Manager to production: a dashboard application launches private OpenStreetMap instances on Kubernetes with Helm, so contributions made in the sandbox go to our own instance instead of the live OpenStreetMap database.
  - Currently building AI features for Tasking Manager in collaboration with fAIr, HOT's AI-assisted mapping platform, integrating its predictions to bring AI-assisted mapping into Tasking Manager.
  - Delivered continuous feature development and fixes, big and small.
  - Reviewed and supervised major features built by external contractors, such as Super Mapper, the Partners module and the MapSwipe integration.
  - Supported the test team by deploying features and fixes to multiple test servers and origins.
  - Engaged with the open source community, handling and addressing the queries it raised.
  - Planned and executed the product roadmap.

- **CultureMap App**
  - A product of Right Place Geo that showcases culture through a map-first approach, rebuilt from a legacy system.
  - Overhauled the entire data model and database with dynamic tables, so organizations can define their own tables without writing new code.
  - Migrated all the legacy data, along with existing Fulcrum field data, into the new tables.
  - Prototyped CultureMap Field, an offline-first field data collection app to replace Fulcrum. It collects data without a connection and syncs it into the organization's tables in batches once the user is back online.
  - Built raster processing and orthomosaic support, overlaying vector layers on top to tell stories, with GeoStyler for styling.
  - Served map layers as PMTiles for fast raster and vector display, and provided analysis URLs as Cloud Optimized GeoTIFFs (COG) for raster and GeoParquet for vector data.
  - Ran processing on AWS: Step Functions trigger the pipeline, Lambda converts the data, and the results are stored per organization in S3.
  - Worked closely with QGIS for analysis, with the CultureMap app providing the analysis URLs for every dataset.
  - Built a QGIS plugin for direct access to all datasets during analysis.
  - Added a live-layer export: a QGIS layer definition (QLR) file, secured with a short-lived token, keeps the styling and loads the layer straight into QGIS.
  - Added raw data downloads, and tabular datasets where uploaded CSVs are converted to Parquet for visualization in Power BI.
- **Ministry of Finance's - Development Finance Information Management System (DFIMS)**
  - Ministry of Finance Nepal's foreign aid information management system.
  - Composite application with Django and postgres for relational maintenance of the complex data and and Fastapi and MongoDB for efficient visualisation of data with complex charts and filters and pandas for report manipulation and generation.
  - The fundings to be visualised with the breakdown to each sectors, locations, development partners etc capable to visualize the data and reports efficiently because of the composite architecture.
- **BIPAD portal - Building Exposure**
  - A Django web application for buildings and facilities mapping, visualization and analysis. Data collection done on feature level with odk conducted for government buildings, schools and offices all over Nepal for risk assessment and hazard analysis.
  - Postgis leveraged spatial queries, proximity analysis, risk analysis, 3D buildings (polygons) visualization.
  - Raster and vector layers for hazard analysis.
  - A risk assessment module of the Nepal's national Bipad Portal developed by National Disaster Risk Reduction and Management Authority (NDRRMA)
- **Geonadir**
  - A cloud-based drone mapping platform.
  - Carried out development of first phase of the product.
  - Drone Imagery Upload, metadata validation
  - Orthophoto generation.

  - Collaborative mapping (WebSocket) using Django Signals.
  - Zapier (web hook) Integration for Slack and data exports.

- **DVS**
  - Information management system for British Embassy Kathmandu (BEK) for project/programs execution, management and monitoring.
  - Spatial queries and vector tiles for demographic data visualisation and filters.
  - Role based management hierarchy.

- **Provincial Climate Change Management Information System (PCCIMS)**
  - Agriculture and climate information management system and knowledge repository for Karnali province of Nepal.
  - GIS vector and raster layers for information dissemination for local levels in Karnali province.

- **Greenway Cycle App**
  - Greenway cycle app available both on apple and play store.
  - Developed the entire django backend for the application.
  - User's activities tracking like distance travelled, carbon saved etc.
  - Django allauth implementation for third party logins.
  - Marketplace implementation.
  - ImePay payment gateway implementation.

## Skills

Languages

- Python (primary)
- JavaScript, HTML, CSS (Basics)

Databases

- SQL (postgres)
- NoSQL (MongoDB)

Frameworks & Libraries

- Django, FastAPI, Flask
- Django REST Framework, SQLAlchemy

GIS & Spatial Data

- PostGIS, handling vector and raster geospatial data

Testing

- Unit and integration testing: `pytest`, Django test framework
- Performance testing: `k6`, monitored with Grafana + InfluxDB

## Skills (continued)

Monitoring & Observability

- New Relic, Sentry, Grafana, InfluxDB

DevOps & Cloud

- AWS service concepts and basic implementation. (EC2, S3, RDS, IAM, Lambda, CloudWatch)
- Deployment concepts and CI/CD with GitHub Actions and IaaC with Terraform.

Project & Collaboration

- Academic and industry research experience
- Project planning and execution
- Effective communication with clients and communities

Frontend

- Good understanding of frontend technologies and integration workflows

## Miscellaneous

- **Blog Post – Migrating Tasking Manager**: Shared technical insights on migrating the Tasking Manager from Flask to FastAPI and from psycopg2 to asyncpg. Covered performance gains, async programming, and maintainability improvements.

- **Hackathon for Field Mapping Task-Splitting Algorithm (Mentor):** Guided participants during the FMTM Hackathon, helping them with GIS-related queries, spatial data issues, and general project support. Provided mentorship and encouraged innovative problem-solving in geospatial applications.

## References

**Ramya Ragupathy**
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